

Worksheet No. - 05

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Subject Name: Advanced Internet Programming Lab **Subject Code:** 25CAP-656

Aim/Overview of the practical: Write a Java Program to implement CRUD (Create, Read, Update, Delete) operations in using MySQL.

Objective: To create a web interface for Student Registration using HTML forms. To implement CRUD operations (Insert, View, Update, Delete) using Java Servlets. To establish database connectivity between a Java web application and MySQL using JDBC.

Procedure/Algorithm/Code

Index.html

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Employee Registration</title>
```

```
</head>
```

```
<body>
```

```
<h2>Insert Employee</h2>
```

```
<form action="EmployeeServlet" method="post">
```

```
<input type="hidden" name="action" value="insert">
```

```
ID: <input type="text" name="employee_id" required><br><br>
```

```
Name: <input type="text" name="name" required><br><br>
```

Age: <input type="number" name="age" required>

Department: <input type="text" name="department" required>

<input type="submit" value="Insert">

</form>

<hr>

<h2>Update Employee</h2>

<form action="EmployeeServlet" method="post">

<input type="hidden" name="action" value="update">

ID: <input type="text" name="employee_id" required>

Age: <input type="number" name="age" required>

Department: <input type="text" name="department" required>

<input type="submit" value="Update">

</form>

<hr>

<h2>Delete Employee</h2>

<form action="EmployeeServlet" method="post">

<input type="hidden" name="action" value="delete">

ID: <input type="text" name="employee_id" required>

<input type="submit" value="Delete">

</form>

<hr>

<h2>View Employees</h2>

<form action="EmployeeServlet" method="get">

<input type="hidden" name="action" value="view">

<input type="submit" value="View All">

</form>

</body>

</html>

Employee.java

```
package com.app;

import java.io.IOException;

import java.io.PrintWriter;

import java.sql.*;

import jakarta.servlet.ServletException;

import jakarta.servlet.annotation.WebServlet;

import jakarta.servlet.http.HttpServlet;

import jakarta.servlet.http.HttpServletRequest;

import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/EmployeeServlet")

public class Employee extends HttpServlet {

    Connection con;

    public void init() {

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            con = DriverManager.getConnection(

                "jdbc:mysql://localhost:3306/employeedb",

                "root",

                "abc@123"

            );

        }

    }

}
```

```
} catch (Exception e) {  
    e.printStackTrace();  
}  
}  
  
protected void doPost(HttpServletRequest req, HttpServletResponse res)  
    throws ServletException, IOException {  
    res.setContentType("text/html");  
    PrintWriter out = res.getWriter();  
    String action = req.getParameter("action");  
    int id = Integer.parseInt(req.getParameter("employee_id"));  
    try {  
        if (action.equals("insert")) {  
            PreparedStatement ps = con.prepareStatement(  
                "INSERT INTO employee VALUES (?, ?, ?, ?)"  
            );  
            ps.setInt(1, id);  
            ps.setString(2, req.getParameter("name"));  
            ps.setInt(3, Integer.parseInt(req.getParameter("age")));  
            ps.setString(4, req.getParameter("department"));  
            ps.executeUpdate();  
            out.println("Employee Inserted Successfully");  
        }  
        if (action.equals("update")) {  
            PreparedStatement ps = con.prepareStatement(  
                "UPDATE employee SET age=?, department=? WHERE employee_id=?"
```

```
);

ps.setInt(1, Integer.parseInt(req.getParameter("age")));

ps.setString(2, req.getParameter("department"));

ps.setInt(3, id);

ps.executeUpdate();

out.println("Employee Updated Successfully");

}

if (action.equals("delete")) {

    PreparedStatement ps = con.prepareStatement(

        "DELETE FROM employee WHERE employee_id=?"

    );

    ps.setInt(1, id);

    ps.executeUpdate();

    out.println("Employee Deleted Successfully");

}

} catch (Exception e) {

    out.println(e);

}

}

protected void doGet(HttpServletRequest req, HttpServletResponse res)

    throws ServletException, IOException {

    res.setContentType("text/html");

    PrintWriter out = res.getWriter();

    try {
```

```
Statement st = con.createStatement();

ResultSet rs = st.executeQuery("SELECT * FROM employee");

out.println("<table border=1>");

out.println("<tr><th>ID</th><th>Name</th><th>Age</th><th>Department</th></tr>");

while (rs.next()) {

    out.println("<tr>");

    out.println("<td>" + rs.getInt(1) + "</td>");

    out.println("<td>" + rs.getString(2) + "</td>");

    out.println("<td>" + rs.getInt(3) + "</td>");

    out.println("<td>" + rs.getString(4) + "</td>");

    out.println("</tr>");

}

out.println("</table>");

} catch (Exception e) {

    out.println(e);

}

}
```

SQL Code:

```
CREATE DATABASE employeedb;

USE employeedb;

DROP TABLE employee;

CREATE TABLE employee (

employee_id INT PRIMARY KEY,

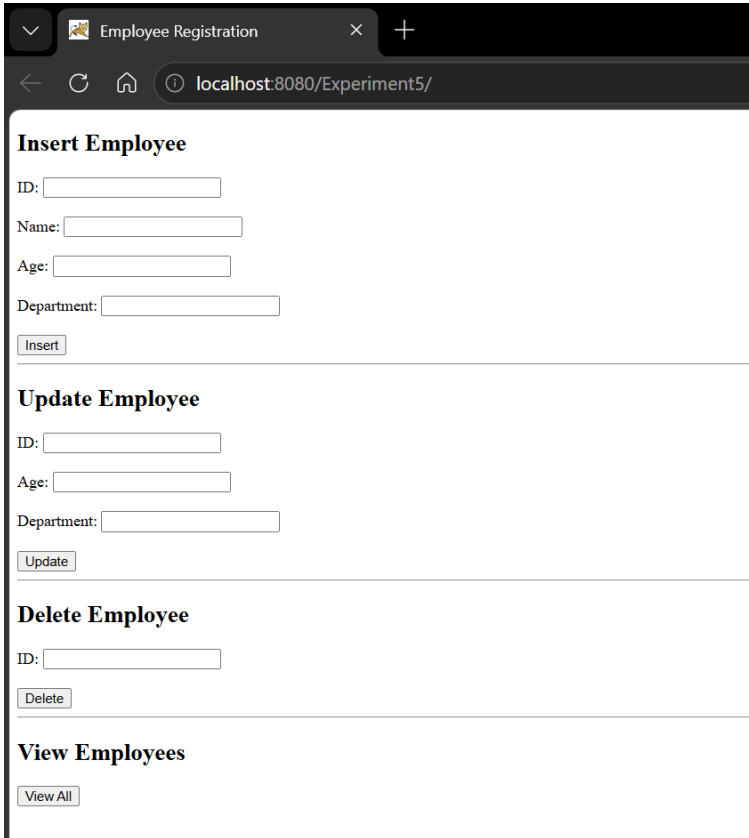
name VARCHAR(100),
```

age INT,

department VARCHAR(100)

);

Output



Employee Registration

localhost:8080/Experiment5/

Insert Employee

ID:

Name:

Age:

Department:

Update Employee

ID:

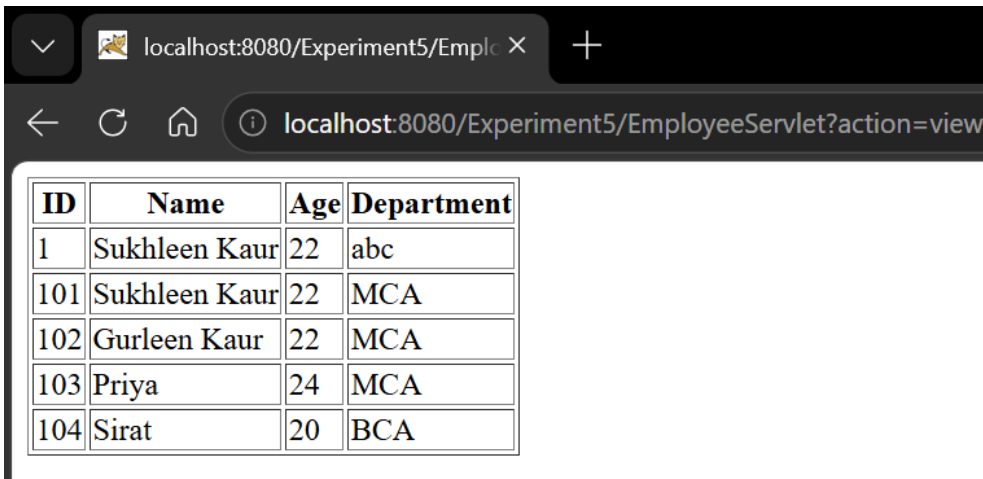
Age:

Department:

Delete Employee

ID:

View Employees



localhost:8080/Experiment5/Emplc

localhost:8080/Experiment5/EmployeeServlet?action=view

ID	Name	Age	Department
1	Sukhleen Kaur	22	abc
101	Sukhleen Kaur	22	MCA
102	Gurleen Kaur	22	MCA
103	Priya	24	MCA
104	Sirat	20	BCA



Learning outcomes (What I have learnt):

1. Understands the working of Java Servlets in web applications.
2. Establish database connection using JDBC and MySQL.
3. Develop a simple dynamic web application using CRUD Operations.
4. Use PreparedStatement to execute SQL queries.